

EXP-014 — Dedicated Document-Level Retrieval

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans · Stage 2 frozen, only document retrieval changed

STATUS — HYPOTHESIS SUPPORTED; FIRST CONFIGURATION TO BEAT THE GLOBAL CONTROL

DOC-C-SECTION reaches **0.875 / 17-of-20** against the global control's 0.775 / 15-of-20 — **2 cases rescued, 0 regressed**. Routing rose from 17/20 to **18/20**, and to **19/20** in a secondary configuration which — notably — retrieves worse.

+0.100
vs global control
2 rescued, 0 regressed

18/20
all documents routed@5
(chunk-derived: 17/20)

0.075
oracle gap remaining
(was 0.175)

310 KB
whole document index
0.18 ms per query

1. Why EXP-014 exists

EXP-012's oracle showed the passage layer is mostly fine — given the right document the same retrievers reach 0.950 / 19-of-20 with zero regressions. EXP-013 then falsified rank aggregation: four rules over chunk rankings all landed on recall@5 0.875 and 17/20. AN-001 explained why — its document contributes one chunk in 300 and the transformer never retrieves it, so no function of those rankings can promote it. **The limitation was the input, not the arithmetic.**

2. What changed, and what did not

Only document retrieval. Stage 2 is byte-identical to EXP-012/013: raw query, full-corpus BM25 plus the existing transformer cosine restricted to selected documents, passage RRF k=60, top 10, statistics never recomputed locally. `top_documents = 5` throughout; same chunks, model (fingerprint `bd95feaeacf98559`), embeddings, anchors and questions.

Every document vector is a deterministic function of embeddings already stored — no new model, no training, no external call. The stored chunk vectors were verified unit-length (14,209 checked, all 1.000) before use. All three reproduction gates pass.

3. Document routing — the primary metric

configuration	@1	@3	@5	@10	routed@5	missing@5
chunk-derived router (EXP-013)	0.525	0.675	0.875	0.925	17/20	AN-001, AN-012, OA-004
DOC-A-MEAN	0.375	0.725	0.825	1.000	16/20	AN-001, AN-003, AN-006, AN-012
DOC-B-CENTROID	0.375	0.725	0.825	1.000	16/20	AN-001, AN-003, AN-006, AN-012
DOC-C-SECTION	0.550	0.875	0.925	0.950	18/20	AN-001, AN-012
DOC-D-MULTIVECTOR	0.500	0.725	0.775	0.925	15/20	AN-001, AN-003, AN-012, OA-002, OA-004
DOC-C-SECTION + chunk	0.600	0.825	0.925	0.925	18/20	AN-001, AN-012
<i>DOC-C-SECTION+chunk+bm25 (secondary)</i>	<i>0.725</i>	<i>0.875</i>	<i>0.950</i>	<i>0.950</i>	<i>19/20</i>	<i>AN-001</i>

4. End-to-end, frozen Stage 2

cell	macro R	full	spans@10	doc R	MRR	absent@300	oracle gap
GLOBAL control	0.775	15/20	17/22	0.925	0.449	2	0.175
DOC-A-MEAN	0.825	16/20	18/22	0.825	0.464	4	0.125
DOC-B-CENTROID	0.825	16/20	18/22	0.825	0.464	4	0.125
DOC-C-SECTION	0.875	17/20	19/22	0.925	0.474	3	0.075
DOC-D-MULTIVECTOR	0.675	13/20	15/22	0.725	0.392	5	0.275
DOC-C-SECTION + chunk	0.875	17/20	19/22	0.925	0.464	3	0.075
<i>DOC-C-SECTION+chunk+bm25 (secondary)</i>	<i>0.825</i>	<i>16/20</i>	<i>18/22</i>	<i>0.950</i>	<i>0.453</i>	<i>2</i>	<i>0.125</i>
ORACLE — not deployable	0.950	19/20	21/22	1.000	0.646	0	0.000

5. What made the difference: one section, one vote

DOC-C differs from DOC-A by a single decision — sections contribute equally instead of chunks — and it is worth two cases of routing and one end-to-end. The corpus explains it: documents average 29.37 sections but the largest has **1,445**. Under a plain chunk mean a document's vector is dominated by whichever section produced the most chunks.

DOC-B is the clean negative. Removing 2,542 exact-duplicate chunks — 18% of the corpus — changed **nothing**: identical routing, identical end-to-end, identical MRR. Duplicate content was not distorting the centroid; uneven section size was.

DOC-D is the clean failure. Scoring a document at its best section is too permissive — any document with one vaguely matching section scores highly — so precision collapses (15/20 routed, document recall 0.725) and it is the only representation that loses to global.

BETTER ROUTING IS NOT BETTER RETRIEVAL

DOC-C-SECTION routes **18/20** and retrieves **0.875 / 17-of-20**. DOC-C-SECTION+chunk+bm25 routes **19/20** — better — and retrieves **0.825 / 16-of-20** — worse.

Adding BM25 buys AN-012's second document but changes which five documents the passage stage competes over, and the net is one case lost. **Document recall is necessary but not sufficient**, which is exactly what the promotion rule was written to catch.

6. AN-001 — improved, still unrouted

ranking	AN-001 document rank
chunk-derived router	never retrieved
DOC-A-MEAN	8
DOC-C-SECTION	13
DOC-D-MULTIVECTOR	74
DOC-C-SECTION+chunk	72
DOC-C-SECTION+chunk+bm25	39

A document-level representation *can* see AN-001's document — DOC-A puts it at rank 8 where every chunk-derived router put it at 62–72 or nowhere, confirming the EXP-013 diagnosis directly. But it is still outside the top 5, and **fusion destroys the gain**: adding the chunk router, which ranks this document nowhere, pushes it back to 72. AN-001 is the one case no EXP-014 configuration routes.

7. AN-012 — multi-hop

ranking	document A	document B	both in top 5?
chunk-derived router	2	30	no
DOC-C-SECTION	1	8	no
DOC-C-SECTION+chunk+bm25	2	5	yes

The only configuration that routes AN-012 fully is the 19/20 secondary one — the same one that retrieves worse overall.

8. Other movements and complementarity

- **OA-004** (long-standing regression watch): chunk router 7 → DOC-C **3**, routed, and not regressed by DOC-C standalone. It is lost by DOC-A+chunk and DOC-B+chunk, so the watch remains warranted.
- **AN-006**: 4 → **2**, rescued end-to-end. **AN-008**: rank 1 everywhere — the case EXP-013's aggregating routers lost is safe here.
- **EXP-013's "BM25 adds +0" is revised**. That was true of chunk-derived routing. Adding BM25 to DOC-C moves routing 18/20 → 19/20 and recall@1 0.550 → 0.725 — they fail differently. It still does not convert end-to-end, so it does not earn a Stage-1 place yet.

9. Reranker gate

cell	1–10	11–30	31–50	51–100	absent	c@30	c@50	c@100
GLOBAL	17	1	1	1	2	0.818	0.864	0.909
DOC-C-SECTION	19	0	0	0	3	0.864	0.864	0.864
DOC-C-SECTION+chunk+bm25	18	2	0	0	2	0.909	0.909	0.909
ORACLE	21	1	0	0	0	1.000	1.000	1.000

DOC-C-SECTION leaves nothing for a reranker — 19 of 22 spans are already in the top 10 and the remaining 3 are absent at 300, so its ceiling is flat at 0.864. A reranker cannot reorder what was never retrieved. The secondary configuration reaches 0.909 at pool 30, which global only reaches at pool 100. Neither justifies building one: the best system's residual is a *retrieval* problem, not a reordering one.

10. Cost

representation	vectors	storage	build	retrieval
DOC-A-MEAN	202	303 KB	0.04 s	0.34 ms
DOC-B-CENTROID	202	303 KB	0.04 s	0.21 ms

representation	vectors	storage	build	retrieval
DOC-C-SECTION	202	303 KB	0.20 s	0.18 ms
DOC-D-MULTIVECTOR	5,933	8900 KB	0.17 s	2.13 ms

Loading the stored chunk vectors takes 1.8 s once. Document-level retrieval is essentially free at this corpus size.

11. Residual failure taxonomy after DOC-C-SECTION

case	classification
AN-001	DOCUMENT REPRESENTATION FAILURE — best rank 8, outside top 5
AN-012	MULTI-HOP ROUTING FAILURE — second document at rank 8
AN-003	WITHIN-DOCUMENT PASSAGE RANKING FAILURE — routed at rank 5, oracle rank 29

Three residual failures, three distinct mechanisms. **None is a passage-conversion failure** — every correctly routed document converted. AN-003 is unchanged and was not tuned around, as declared.

12. Limitations

- $n = 20 / 22$ spans; one case is 5 percentage points. **No significance claims** — DOC-C's +2 cases is a two-case difference on twenty questions.
- Four representations, preregistered, run once each. No variant was selected after seeing results.
- DOC-C's advantage rests on one design decision, well-motivated by the 1,445-section outlier but evidenced on a single corpus.
- **EXP-NUL** remains **BLOCKED** — still no measured no-retrieval floor.

13. Did document representation solve routing?

Partly — and it did something better: it converted. Routing went 17/20 → 18/20 standalone and 19/20 with BM25, short of solving it. But for the first time an intervention **beat the global control end-to-end**, with zero regressions, closing the oracle gap from 0.175 to 0.075. AN-001 moving from *never retrieved* to rank 8 is the EXP-013 diagnosis confirmed in one number.

PROMOTION ASSESSMENT

The preregistered bar was routed@5 $\geq 19/20$ **and** end-to-end > 0.775 **and** limited/no regressions. DOC-C-SECTION meets two of three — 0.875 / 17-of-20 with zero regressions — but routes 18/20, and the configuration that does route 19/20 retrieves worse. **The formal bar is not met and it is not promoted.** It is nonetheless the strongest candidate this project has produced.

14. What the measurements justify next

1. **Replicate DOC-C-SECTION before promoting anything** — a +2-case result at $n=20$ needs confirmation, ideally on held-out questions rather than this development set.
2. **Attack AN-001 as a representation problem.** It is the single remaining routing failure, visible at rank 8 to a plain mean, and fusion is what buries it.

3. **Do not add BM25 to Stage 1 yet** — it improves routing and costs retrieval.
4. **A reranker is still not justified** — the residual is 3 spans absent at 300.
5. **The golden set is now the binding constraint on inference.** Six of the last seven experiments turned on one or two cases; expanding it would buy more than any further retrieval change.

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